

Foundations of TensorFlow

[TensorFlow]

1.0 Content Block

Metrics In order to assess the performance of machine learning models, TensorFlow gives API access to commonly used metrics. Examples include various accuracy metrics (binary, categorical, sparse categorical) along with other metrics such as Precision, Recall, and Intersection-over-Union (IoU).

2.0 Content Block

Google Colab Google also released Collaboratory, a TensorFlow Jupyter notebook environment that does not require any setup. It runs on Google Cloud and allows users free access to GPUs and the ability to store and share notebooks on Google Drive.

3.0 Content Block

TensorFlow TensorFlow is Google Brain's second-generation system. Version 1.0.0 was released on February 11, 2017. While the reference implementation runs on single devices, TensorFlow can run on multiple CPUs and GPUs (with optional CUDA and SYCL extensions for general-purpose computing on graphics processing units). TensorFlow is available on 64-bit Linux, macOS, Windows, and mobile computing platforms including Android and iOS. Its flexible architecture allows for easy deployment of computation across a variety of platforms (CPUs, GPUs, TPUs), and from desktops to clusters of servers to mobile and edge devices. TensorFlow computations are expressed as stateful dataflow graphs. The name TensorFlow derives from the operations that such neural networks perform on multidimensional data arrays, which are referred to as tensors. During the Google I/O Conference in June 2016, Jeff Dean stated that 1,500 repositories on GitHub mentioned TensorFlow, of which only 5 were from Google. In March 2018, Google announced TensorFlow.js version 1.0 for machine learning in JavaScript. In Jan 2019, Google announced TensorFlow 2.0. It became officially available in September 2019. In May 2019, Google announced TensorFlow Graphics for deep learning in computer graphics.

4.0 Content Block

Medical GE Healthcare used TensorFlow to increase the speed and accuracy of MRIs in identifying specific body parts. Google used TensorFlow to create DermAssist, a free mobile application that allows users to take pictures of their skin and identify potential health complications. Sinovation Ventures used TensorFlow to identify and classify eye diseases from optical coherence tomography (OCT) scans.

5.0 Content Block

Extensions TensorFlow also offers a variety of libraries and extensions to advance and extend the models and methods used. For example, TensorFlow Recommenders and TensorFlow Graphics are libraries for their respective functional. Other add-ons, libraries, and frameworks include TensorFlow Model Optimization, TensorFlow Probability, TensorFlow Quantum, and TensorFlow Decision Forests.

6.0 Content Block

TensorFlow Lite In May 2017, Google announced TensorFlow Lite as a software stack to support machine learning models for mobile and embedded devices, and in November 2017, provided the developer preview. In January 2019, the TensorFlow team released a developer preview of the mobile GPU inference engine with OpenGL ES 3.1 Compute Shaders on Android devices and Metal Compute Shaders on iOS devices. In May 2019, Google announced that their TensorFlow Lite Micro (also known as TensorFlow Lite for Microcontrollers) and ARM's uTensor would be merging. It was renamed as LiteRT in

2024.

7.0 Content Block

TensorFlow.js TensorFlow also has a library for machine learning in JavaScript. Using the provided JavaScript APIs, TensorFlow.js allows users to use either Tensorflow.js models or converted models from TensorFlow or TFLite, retrain the given models, and run on the web.

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LiteRT LiteRT, formerly known as TensorFlow Lite, has APIs for mobile apps or embedded devices to generate and deploy TensorFlow models. These models are compressed and optimized in order to be more efficient and have a higher performance on smaller capacity devices. LiteRT uses FlatBuffers as the data serialization format for network models, eschewing the Protocol Buffers format used by standard TensorFlow models.

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TF.nn TensorFlow.nn is a module for executing primitive neural network operations on models. Some of these operations include variations of convolutions (1/2/3D, Atrous, depthwise), activation functions (Softmax, RELU, GELU, Sigmoid, etc.) and their variations, and other operations (max-pooling, bias-add, etc.).

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In May 2016, Google announced its Tensor processing unit (TPU), an application-specific integrated circuit (ASIC, a hardware chip) built specifically for machine learning and tailored for TensorFlow. A TPU is a programmable AI accelerator designed to provide high throughput of low-precision arithmetic (e.g., 8-bit), and oriented toward using or running models rather than training them. Google announced they had been running TPUs inside their data centers for more than a year, and had found them to deliver an order of magnitude better-optimized performance per watt for machine learning. In May 2017, Google announced the second-generation, as well as the availability of the TPUs in Google Compute Engine. The second-generation TPUs deliver up to 180 teraflops of performance, and when organized into clusters of 64 TPUs, provide up to 11.5 petaflops. In May 2018, Google announced the third-generation TPUs delivering up to 420 teraflops of performance and 128 GB high bandwidth memory (HBM). Cloud TPU v3 Pods offer 100+ petaflops of performance and 32 TB HBM. In February 2018, Google announced that they were making TPUs available in beta on the Google Cloud Platform.

11.0 Content Block

TFX TensorFlow Extended (abbrev. TFX) provides numerous components to perform all the operations needed for end-to-end production. Components include loading, validating, and transforming data, tuning, training, and evaluating the machine learning model, and pushing the model itself into production.

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Retail The e-commerce platform Carousell used TensorFlow to provide personalized recommendations for customers. The cosmetics company ModiFace used TensorFlow to create an augmented reality experience for customers to test various shades of make-up on their face.

13.0 Content Block

TensorFlow TensorFlow serves as a core platform and library for machine learning. TensorFlow's APIs use Keras to allow users to make their own machine-learning models. In addition to building and training their model, TensorFlow can also help load the data to train the model, and deploy it using TensorFlow Serving. TensorFlow provides a stable Python Application Program Interface (API), as well as APIs without backwards compatibility guarantee for JavaScript, C++, and Java. Third-party language binding

packages are also available for C#, Haskell, Julia, MATLAB, Object Pascal, R, Scala, Rust, OCaml, and Crystal. Bindings that are now archived and unsupported include Go and Swift.

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Edge TPU In July 2018, the Edge TPU was announced. Edge TPU is Google's purpose-built ASIC chip designed to run TensorFlow Lite machine learning (ML) models on small client computing devices such as smartphones known as edge computing.